Regional Quantitative Problem-Solving Circle

Module 6: Mathematical Modeling & Synthesis (Weeks 11–12)

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1 Axiomatic & Theoretical Foundation

Many discrete combinatorial problems are equivalent to tracking a particle moving randomly across a finite set of states. By mapping these systems onto directed graphs, we transition from ad hoc counting arguments to robust linear algebra and transition matrices.

Definition 1 (Discrete-Time Markov Chain)**.** A discrete-time Markov chain is a sequence of random variables $X_0, X_1, X_2, \dots$ taking values in a finite state space S, which satisfies the memoryless property:

$$\mathbb{P}(X_{n+1} = j \mid X_n = i, X_{n-1} = i_{n-1}, \dots, X_0 = i_0) = \mathbb{P}(X_{n+1} = j \mid X_n = i) = p_{ij}$$

The future state depends exclusively on the current state, completely independent of the past history.

Definition 2 (Absorbing State & Hitting Time)**.** A state $k \in S$ is absorbing if $p_{kk} = 1$ and $p_{kj} = 0$ for all $j \neq k$. The expected hitting time E_i is the expected number of transitions to reach an absorbing state, given that the process starts in state i.

2 The Core Invariant / State Function

To calculate expected values in a Markov chain, we deploy **First-Step Analysis**.
Core Invariant Framework: The expected time to reach an absorbing state from state i is exactly 1 transition (the step we are about to take), plus the weighted sum of the expected hitting times from whatever state j we land in.

$$E_i = 1 + \sum_{j \in S} p_{ij} E_j$$

This creates a deterministic system of linear equations mapping the invariant stochastic topology of the state space.

3 Lemma & Canonical Proof

We utilize First-Step Analysis to derive hitting times on a highly symmetric cyclic graph, circumventing the need for infinite summation of path probabilities.

Lemma 1 (Cyclic Random Walk Hitting Times)**.** *A particle moves on a discrete cycle of* 5 *nodes, labeled* $0, 1, 2, 3, 4$ *in clockwise order. At each step, it moves one node clockwise with probability* $p = 0.5$ *or one node counter-clockwise with probability* $q = 0.5$*. If node* 0 *is an absorbing state, find the expected number of steps to reach node* 0 *starting from node* 1*.*

Proof. Let E_k be the expected number of steps to reach the absorbing state 0 given that the particle is currently at node k. By definition of an absorbing state, if the particle starts at node 0, it requires zero steps to reach it:

$$E_0 = 0$$

For the remaining nodes $k \in \{1, 2, 3, 4\}$, we apply the First-Step Analysis invariant. From node k, the particle takes exactly 1 step. It then lands on node $k - 1 \pmod 5$ with probability 0.5, or node $k + 1 \pmod 5$ with probability 0.5. This generates a system of linear recurrences:

$$\begin{aligned}
E_1 &= 1 + 0.5E_0 + 0.5E_2 \\
E_2 &= 1 + 0.5E_1 + 0.5E_3 \\
E_3 &= 1 + 0.5E_2 + 0.5E_4 \\
E_4 &= 1 + 0.5E_3 + 0.5E_0
\end{aligned}$$

Substitute the boundary condition $E_0 = 0$ into the equations for E_1 and E_4:

$$E_1 = 1 + 0.5E_2 \tag{1}$$
$$E_2 = 1 + 0.5E_1 + 0.5E_3 \tag{2}$$
$$E_3 = 1 + 0.5E_2 + 0.5E_4 \tag{3}$$
$$E_4 = 1 + 0.5E_3 \tag{4}$$

Due to the topological symmetry of the cycle and the unbiased probabilities (0.5 in both directions), the expected distance from node 1 to node 0 is identical to the expected distance from node 4 to node 0. Similarly, the distance from node 2 equals the distance from node 3. Therefore, we can assert $E_1 = E_4$ and $E_2 = E_3$.

Substituting $E_2 = E_3$ into equation (2):

$$E_2 = 1 + 0.5E_1 + 0.5E_2 \implies 0.5E_2 = 1 + 0.5E_1 \implies E_2 = 2 + E_1$$

Now substitute this expression for E_2 into equation (1):

$$E_1 = 1 + 0.5(2 + E_1) = 1 + 1 + 0.5E_1 = 2 + 0.5E_1$$

Subtract $0.5E_1$ from both sides:

$$0.5E_1 = 2 \implies E_1 = 4$$

We can also find E_2:

$$E_2 = 2 + 4 = 6$$

Thus, the expected number of steps to reach the absorbing state from node 1 is exactly $E_1 = 4$. $\square$

4 Seminar Heuristics & Socratic Framework

During the whiteboard defense phase, push students on the following diagnostic cues and common pitfalls:

- **Diagnostic Cue — "When to apply?":** Whenever a stochastic process has "memoryless" transitions (like coin flips, die rolls, or unbiased graph traversal) and asks for "expected steps" or "probability of reaching state A before state B", immediately mandate drawing a finite state machine and applying First-Step Analysis.
- **Pitfall — The Infinite Series Trap:** Students will instinctively try to write out an infinite geometric series for every possible path to the end state. **Socratic Prompt:** *"What happens to your summation if the particle loops between node 2 and node 3 a billion times? Do you want to calculate all those paths, or can we just define an invariant state variable E_2?"*
- **Pitfall — Boundary Conditions:** Students routinely forget to explicitly set the expected time of the absorbing state to zero. **Socratic Prompt:** *"If you are already standing at the destination, how many steps does it take to arrive?"*